

Smaran Rangarajan Bharadwaj

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Python3 • Go • C • C++ • Java
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EDUCATION

- **RV University, Bengaluru** • B.Tech (Hons.), Computer Science and Engineering Graduating: Aug 2026 • CGPA: 9.80
- **Sri Kumaran Children's Home, Bengaluru** • Class 12 (CBSE) - PCMC 2022 • Score: 95.2%

TECHNICAL SKILLS

Developer Tools: Docker, Kubernetes, Git, AWS, GCP, Azure, Jira, Confluence, HuggingFace
Databases: MySQL, SQLite, MongoDB, ClickHouse
Frameworks: TensorFlow, PyTorch, YOLO, Flask, Streamlit, Ollama, Kafka, Flink, Unsloth, gRPC, RESTful APIs

EXPERIENCE

Software Development Intern Jan 2026 - Jul 2026
Aviatrix Systems Pvt. Ltd. Bengaluru

- Designed and shipped the scalable Data-Bridge backend in Go using gRPC and ClickHouse materialized views, enabling sub-second queries across 100,000+ records.
- Architected and shipped a distributed multi-cloud (AWS/GCP) risk assessment pipeline leveraging Flink and ClickHouse) analyzing 22 resource types in under 100 seconds.
- Automated the CI/CD workflows for the cloud-native Data-Bridge service with GitHub Actions, reducing deployment time by 68%.

ML Project Intern Aug 2025 - Jan 2026
Aeronautical Development Establishment (DRDO) Bengaluru

- Architected a multi-stage mission debriefing pipeline processing 10,000+ flight parameters across 6 missions in under 15 minutes with parallelized processing.
- Engineered an event detection engine using Allen's Interval Algebra and expert systems logic to classify overlapping continuous and discrete parameter anomalies.

ML Engineer Intern Jun 2025 - Aug 2025
GovernAIce Berkeley (Remote)

- Designed an end-to-end multi-lingual RAG pipeline reducing compliance report generation time by 73% over manual processes by leveraging DeepSeek-R1:70B and MongoDB Atlas Vector Search.
- Developed a policy ingestion pipeline supporting 6 languages and 12,000+ legal documents utilizing IBM Docling and XLM-RoBERTa embeddings.

LEADERSHIP AND AWARDS

- Founded and established RV University's IEEE Student Branch, growing it from 5 founding ExeCom members to 60+ members across 3 society chapters and 20+ events.
- Awarded merit scholarships for 3 consecutive years, ranking 2nd among the university's top academic performers.
- Won 2nd place among 117 teams at the HPCC Safe Heavens Hackathon for developing a distributed computing solution for refugee support systems.
- Co-designed and published a patent for Gelos, a fitness-focused interactive game, alongside Dr. Phani Kumar Pullela.

PROJECTS

Efficient and Interpretable Deep Learning Framework for PCOS Detection Neural Architecture Search, XAI

- Architected a DARTS-based Neural Architecture Search pipeline discovering an optimized CNN for PCOS ultrasound classification across 1,924 images, independently achieving 97.91% test accuracy.
- Integrated Explainable AI, zero-shot, and few-shot learning methods, retaining 94.72% accuracy in limited-data scenarios.

Braille GPT Multi-agent System, LoRA

- Developed a multi-agent system built on LoRA-fine-tuned LLaMA 3.1 8B with a Streamlit-based front end, converting text and image inputs into bilingual Braille with a 0.76 average BLEU score across translation agents.

Obstacle Avoidance for Visually Impaired Computer Vision, YOLO, Transfer Learning

- Led a four-member team to develop a YOLOv8n-based obstacle detection system for the visually impaired, achieving 82.4% mAP on a self-collected dataset of 500+ pedestrian images.
- Built and integrated a real-time audio alert pipeline, achieving 132ms average inference latency when deployed on edge.